

Not One Kind of Force

Name: _____

Three investigations, three laws, and a shared habit of notation that quietly asks students to treat all three as the same kind of thing.

Coulomb
 $F(r)$

Lorentz
 $F(v)$

Ohm
 $F(\text{material})$

The question. Three laws, each written in roughly the shape “force equals something”.

Are they the same kind of claim?

Working rule: Completing the comparison table is the investigation. A correct table with no justification earns less than an incomplete table with a defended reason for every entry.

1. Three, side by side

complete the table from memory or derivation

	Coulomb	Lorentz	Ohm
depends on			
source			
field			
type			

2. Whether it does work

conservative, incapable of work, or dissipative

Match each description to the correct force. Each is used exactly once.

Conservative, has a potential energy	_____
Never does any work at all	_____
Dissipative, converts energy to heat	_____

Why is “incapable of work” a stronger and more specific claim than “non-conservative”?

3. Interaction law, or constitutive law

the deepest split

Define each term in your own words, then classify Coulomb, Lorentz, and Ohm.

interaction law

constitutive law

4. More laws to sort

test the classification elsewhere

For each law, decide interaction or constitutive, and justify in one line.

Law	Classification	Reason
Newton's gravitation		
Hooke's law		
Kinetic friction, $F = \mu N$		
Archimedes' principle		

5. One symbol, three claimswhy they all get called F

Why might writing every one of these as “force equals something”, in the same Newton's-second-law grammar, be convenient? What does that convenience risk?

State, in a single sentence, the difference between a relationship that exists independently of any material and one that describes how a particular material happens to respond.

6. Core results

what this enquiry actually earned

The completed comparison, restated.

	Coulomb	Lorentz	Ohm
type	interaction	interaction	constitutive

Concepts earned, in order.

- Two of the three laws depend on where things are or how they move; the third depends on what something is made of.
- Conservative, incapable of work, and dissipative are three distinct relationships with energy, not three points on one scale.
- Interaction laws are universal between any two objects; constitutive laws describe one material's empirical response.
- A shared symbol, F , invites the belief that structurally different claims are the same kind of entity.

Every substantial claim should be accompanied by a reason, a named example, or a counterexample.